File: Customer.java

```java
package ONTAP_GK;

public class Customer {
    private int id;
    private String name;
    private String gender; // "Nam" hoặc "Nu"
    private int age;
    private String blood;  // "A", "B", "AB", "O"

    public Customer(int id, String name, String gender, int age, String blood) {
        this.id = id;
        this.name = name;
        this.gender = gender;
        this.age = age;
        this.blood = blood;
    }

    public int getId() { return id; }
    public String getName() { return name; }
    public String getGender() { return gender; }
    public int getAge() { return age; }
    public String getBlood() { return blood; }

    // Lấy tên chính (từ cuối cùng của họ và tên) phục vụ sắp xếp
    public String getFirstName() {
        String trimmed = name.trim();
        int lastSpaceIndex = trimmed.lastIndexOf(" ");
        if (lastSpaceIndex == -1) {
            return trimmed;
        }
        return trimmed.substring(lastSpaceIndex + 1);
    }

    // In một dòng định dạng bảng
    public void printRow() {
        System.out.printf("| %-8d | %-25s | %-8s | %-5d | %-9s |\n", id, name, gender, age, blood);
    }

    // Chuyển đổi tượng thành định dạng CSV để lưu file
    public String toFileString() {
        return id + "," + name + "," + gender + "," + age + "," + blood;
    }
}
// ==================== KẾT THÚC FILE CUSTOMER.JAVA ====================
```

File: Node.java (Toàn bộ)

```java
package ONTAP_GK;

public class Node {
    Customer data;
    Node left;
    Node right;

    public Node(Customer data) {
        this.data = data;
        this.left = null;
        this.right = null;
    }
}
// ==================== KẾT THÚC FILE NODE.JAVA ====================
```

File: CustomerBST.java (Phần 1: Khai báo, Constructor, Thêm Node)

```java
package ONTAP_GK;

import java.io.*;
import java.util.*;

public class CustomerBST {
    Node root;

    public CustomerBST() {
        root = null;
    }

    // Chèn 1 node vào cây BST so sánh theo customer_ID
    public void insert(Customer customer) {
        root = insertRec(root, customer);
    }

    private Node insertRec(Node current, Customer customer) {
        if (current == null) {
            return new Node(customer);
        }
        if (customer.getId() < current.data.getId()) {
            current.left = insertRec(current.left, customer);
        } else if (customer.getId() > current.data.getId()) {
            current.right = insertRec(current.right, customer);
        } else {
            System.out.println("(!) ID " + customer.getId() + " da ton tai, khong the them.");
        }
        return current;
    }
```

File: CustomerBST.java (Phần 2: YÊU CẦU 1 - Bắt đầu Nhập dữ liệu)

```
    // --- YÊU CẦU 1: Nhập 1 danh sách khách hàng từ bàn phím vào cây BST ---
    public void inputCustomerList(Scanner scanner) {
        int n = 0;
        while (true) {
            try {
                System.out.print("Nhap so luong khach hang can them vao danh sach: ");
                n = Integer.parseInt(scanner.nextLine().trim());
                if (n <= 0) {
                    System.out.println("So luong phai lon hon 0. Vui long nhap lai!");
                    continue;
                }
                break;
            } catch (NumberFormatException e) {
                System.out.println("Vui long nhap mot so nguyen hop le!");
            }
        }

        for (int i = 1; i <= n; i++) {
            System.out.println("\n--- Nhap khach hang thu " + i + "/" + n + " ---");
            int id;
            while (true) {
                try {
                    System.out.print("Nhap ID: ");
                    id = Integer.parseInt(scanner.nextLine().trim());
                    if (searchById(id) != null) {
                        System.out.println("(!) ID " + id + " da ton tai trong cay BST. Vui long nhap ID khac!");
                        continue;
                    }
                    break;
                } catch (NumberFormatException e) {
                    System.out.println("ID phai la so nguyen!");
                }
            }

            System.out.print("Nhap Ho va Ten: ");
            String name = scanner.nextLine().trim();
```

File: CustomerBST.java (Phần 3: YÊU CẦU 1 - Hoàn tất Nhập liệu)

```
            String gender;
            while (true) {
                System.out.print("Nhap Gioi tinh (Nam/Nu): ");
                gender = scanner.nextLine().trim();
                if (gender.equalsIgnoreCase("Nam") || gender.equalsIgnoreCase("Nu")) {
                    gender = gender.equalsIgnoreCase("Nam") ? "Nam" : "Nu";
                    break;
                }
                System.out.println("Gioi tinh chi duoc nhap 'Nam' hoac 'Nu'!");
            }

            int age;
            while (true) {
                try {
                    System.out.print("Nhap Tuoi: ");
                    age = Integer.parseInt(scanner.nextLine().trim());
                    if (age <= 0 || age > 150) {
                        System.out.println("Tuoi phai tu 1 den 150!");
                        continue;
                    }
                    break;
                } catch (NumberFormatException e) {
                    System.out.println("Tuoi phai la so nguyen!");
                }
            }

            String blood;
            while (true) {
                System.out.print("Nhap Nhom mau (A/B/AB/O): ");
                blood = scanner.nextLine().trim().toUpperCase();
                if (blood.equals("A") || blood.equals("B") || blood.equals("AB") || blood.equals("O")) {
                    break;
                }
                System.out.println("Nhom mau khong hop le! Chi chap nhan: A, B, AB, O.");
            }

            insert(new Customer(id, name, gender, age, blood));
            System.out.println("-> Da them khach hang ID " + id + " vao cay BST.");
        }
        System.out.println("\n=> DA HOAN TAT NHAP " + n + " KHACH HANG VAO CAY BST!");
    }
```

File: CustomerBST.java (Phần 4: Duyệt In-Order, In bảng & YÊU CẦU 2)

```java
    // Hàm phụ trợ: Lấy danh sách khách hàng bằng duyệt In-Order (L-N-R)
    public List<Customer> getAllCustomers() {
        List<Customer> list = new ArrayList<>();
        inOrderTraversal(root, list);
        return list;
    }

    private void inOrderTraversal(Node current, List<Customer> list) {
        if (current != null) {
            inOrderTraversal(current.left, list);
            list.add(current.data);
            inOrderTraversal(current.right, list);
        }
    }

    public static void printTableHeader() {
        System.out.println("+----------+--------------------------+----------+-------+-----------+");
        System.out.printf("| %-8s | %-25s | %-8s | %-5s | %-9s |\n", "ID", "Ho va Ten", "Gioi tinh", "Tuoi", "Nhom mau");
        System.out.println("+----------+--------------------------+----------+-------+-----------+");
    }

    public static void printTableFooter() {
        System.out.println("+----------+--------------------------+----------+-------+-----------+");
    }

    // --- YÊU CẦU 2: Xuất danh sách theo định dạng bảng ---
    public void displayAll() {
        List<Customer> list = getAllCustomers();
        if (list.isEmpty()) {
            System.out.println("Danh sach khach hang dang trong!");
            return;
        }
        printTableHeader();
        for (Customer c : list) {
            c.printRow();
        }
        printTableFooter();
    }
```

File: CustomerBST.java (Phần 5: YÊU CẦU 3 & YÊU CẦU 4)

```
    // --- YÊU CẦU 3: Tuổi trung bình, khách hàng lớn tuổi nhất, nhỏ tuổi nhất ---
    public void calculateAgeStats() {
        List<Customer> list = getAllCustomers();
        if (list.isEmpty()) {
            System.out.println("Danh sach trong, khong co du lieu thong ke!");
            return;
        }
        int totalAge = 0;
        Customer oldest = list.get(0);
        Customer youngest = list.get(0);
        for (Customer c : list) {
            totalAge += c.getAge();
            if (c.getAge() > oldest.getAge()) oldest = c;
            if (c.getAge() < youngest.getAge()) youngest = c;
        }
        double avgAge = (double) totalAge / list.size();
        System.out.printf("- Tuoi trung binh cua khach hang: %.2f\n", avgAge);
        System.out.println("- Khach hang lon tuoi nhat (" + oldest.getAge() + " tuoi): " + oldest.getName() + " (ID: " + oldest.getId() + ")");
        System.out.println("- Khach hang nho tuoi nhat (" + youngest.getAge() + " tuoi): " + youngest.getName() + " (ID: " + youngest.getId() + ")");
    }

    // --- YÊU CẦU 4: Xuất danh sách theo giới tính & nhóm máu ---
    public void filterCustomers() {
        List<Customer> list = getAllCustomers();
        if (list.isEmpty()) {
            System.out.println("Danh sach trong!");
            return;
        }
        System.out.println("\n--- DANH SACH KHACH HANG NAM ---");
        printFilteredList(list, c -> c.getGender().equalsIgnoreCase("Nam"));

        System.out.println("\n--- DANH SACH KHACH HANG NU ---");
        printFilteredList(list, c -> c.getGender().equalsIgnoreCase("Nu"));

        System.out.println("\n--- DANH SACH KHACH HANG NHOM MAU A ---");
        printFilteredList(list, c -> c.getBlood().equalsIgnoreCase("A"));

        System.out.println("\n--- DANH SACH KHACH HANG NHOM MAU B ---");
        printFilteredList(list, c -> c.getBlood().equalsIgnoreCase("B"));

        System.out.println("\n--- DANH SACH KHACH HANG NHOM MAU O ---");
        printFilteredList(list, c -> c.getBlood().equalsIgnoreCase("O"));
    }
```

File: CustomerBST.java (Phần 6: Hàm lọc Lambda & YÊU CẦU 5)

```
    private interface FilterCondition {
        boolean test(Customer c);
    }

    private void printFilteredList(List<Customer> list, FilterCondition condition) {
        boolean found = false;
        printTableHeader();
        for (Customer c : list) {
            if (condition.test(c)) {
                c.printRow();
                found = true;
            }
        }
        printTableFooter();
        if (!found) {
            System.out.println("(Khong co khach hang thoa man)");
        }
    }

    // --- YÊU CẦU 5: Sắp xếp theo tên và xuất danh sách khách hàng nam ---
    public void sortAndDisplayMaleCustomers() {
        List<Customer> males = new ArrayList<>();
        for (Customer c : getAllCustomers()) {
            if (c.getGender().equalsIgnoreCase("Nam")) {
                males.add(c);
            }
        }
        if (males.isEmpty()) {
            System.out.println("Khong co khach hang nam nao trong danh sach!");
            return;
        }
        males.sort((c1, c2) -> c1.getFirstName().compareToIgnoreCase(c2.getFirstName()));
        System.out.println("\n=== DANH SACH KHACH HANG NAM DA SAP XEP THEO TEN ===");
        printTableHeader();
        for (Customer c : males) {
            c.printRow();
        }
        printTableFooter();
    }
```

File: CustomerBST.java (Phần 7: YÊU CẦU 6 & YÊU CẦU 7)

```java
    // --- YÊU CẦU 6: Khách hàng nam lớn tuổi nhất, nhỏ tuổi nhất ---
    public void findOldestAndYoungestMale() {
        Customer oldestMale = null;
        Customer youngestMale = null;
        for (Customer c : getAllCustomers()) {
            if (c.getGender().equalsIgnoreCase("Nam")) {
                if (oldestMale == null || c.getAge() > oldestMale.getAge()) {
                    oldestMale = c;
                }
                if (youngestMale == null || c.getAge() < youngestMale.getAge()) {
                    youngestMale = c;
                }
            }
        }
        if (oldestMale == null) {
            System.out.println("Khong co khach hang nam trong danh sach!");
            return;
        }
        System.out.println("\n=== THONG KE KHACH HANG NAM THEO TUOI ===");
        System.out.println("- Khach hang nam lon tuoi nhat: " + oldestMale.getName() + " (" + oldestMale.getAge() + " tuoi, ID: " + oldestMale.getId() +
        System.out.println("- Khach hang nam nho tuoi nhat: " + youngestMale.getName() + " (" + youngestMale.getAge() + " tuoi, ID: " + youngestMale.getI
    }

    // --- YÊU CẦU 7: Tìm kiếm theo ID ---
    public Customer searchById(int id) {
        return searchRec(root, id);
    }

    private Customer searchRec(Node current, int id) {
        if (current == null) {
            return null;
        }
        if (id == current.data.getId()) {
            return current.data;
        }
        if (id < current.data.getId()) {
            return searchRec(current.left, id);
        }
        return searchRec(current.right, id);
    }
```

File: CustomerBST.java (Phần 8: YÊU CẦU 8 - Xóa Node trên cây BST)

```
    // --- YÊU CẦU 8: Xóa theo ID ---
    public boolean deleteById(int id) {
        if (searchById(id) == null) {
            return false;
        }
        root = deleteRec(root, id);
        return true;
    }

    private Node deleteRec(Node current, int id) {
        if (current == null) return null;
        if (id < current.data.getId()) {
            current.left = deleteRec(current.left, id);
        } else if (id > current.data.getId()) {
            current.right = deleteRec(current.right, id);
        } else {
            // Trường hợp node lá hoặc có 1 con
            if (current.left == null) return current.right;
            if (current.right == null) return current.left;
            // Trường hợp node có 2 con: tìm node nhỏ nhất bên cây con phải
            Node minNode = findMin(current.right);
            current.data = minNode.data;
            current.right = deleteRec(current.right, minNode.data.getId());
        }
        return current;
    }

    private Node findMin(Node node) {
        while (node.left != null) {
            node = node.left;
        }
        return node;
    }
```

File: CustomerBST.java (Phần 9: YÊU CẦU 9 - Đọc/Ghi file CSV & Hoàn tất Class)

```
    // --- YÊU CẦU 9: Lưu file và Đọc file ---
    public void saveToFile(String filename) {
        List<Customer> list = getAllCustomers();
        try (BufferedWriter bw = new BufferedWriter(new FileWriter(filename))) {
            for (Customer c : list) {
                bw.write(c.toFileString());
                bw.newLine();
            }
            System.out.println("-> Da luu thanh cong " + list.size() + " khach hang vao file: " + filename);
        } catch (IOException e) {
            System.out.println("Loi khi luu file: " + e.getMessage());
        }
    }

    public void loadFromFile(String filename) {
        File file = new File(filename);
        if (!file.exists()) {
            System.out.println("(!) File '" + filename + "' chua ton tai. Chuong trinh khoi tao cay rong.");
            return;
        }
        try (BufferedReader br = new BufferedReader(new FileReader(filename))) {
            String line;
            int count = 0;
            while ((line = br.readLine()) != null) {
                line = line.trim();
                if (line.isEmpty()) continue;
                String[] parts = line.split(",");
                if (parts.length == 5) {
                    int id = Integer.parseInt(parts[0].trim());
                    String name = parts[1].trim();
                    String gender = parts[2].trim();
                    int age = Integer.parseInt(parts[3].trim());
                    String blood = parts[4].trim();
                    insert(new Customer(id, name, gender, age, blood));
                    count++;
                }
            }
            System.out.println("-> Da doc thanh cong " + count + " khach hang tu file: " + filename);
        } catch (Exception e) {
            System.out.println("Loi khi doc file: " + e.getMessage());
        }
    }
}
// ==================== KẾT THÚC FILE CUSTOMERBST.JAVA ====================
```

File: Main.java (Phần 1: Menu & Xử lý Case 1 - 4)

```java
package ONTAP_GK;

import java.util.Scanner;

public class Main {
    private static final String FILE_NAME = "customers.txt";

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        CustomerBST bst = new CustomerBST();

        // Nạp dữ liệu từ file khi khởi động nếu file đã tồn tại
        System.out.println("=== KHOI DONG HE THONG QUAN LY KHACH HANG ===");
        bst.loadFromFile(FILE_NAME);

        int choice = -1;
        do {
            System.out.println("\n============== MENU QUAN LY BST CUSTOMER ==============");
            System.out.println("1. Nhap 1 danh sach khach hang vao cay BST");
            System.out.println("2. Xuat danh sach khach hang (dinh dang bang)");
            System.out.println("3. Tinh tuoi TB, tim khach hang lon tuoi / nho tuoi nhat");
            System.out.println("4. Xuat DS theo gioi tinh (Nam, Nu) va nhom mau (A, B, O)");
            System.out.println("5. Sap xep va xuat danh sach khach hang Nam theo ten");
            System.out.println("6. Tim khach hang Nam lon tuoi nhat & nho tuoi nhat");
            System.out.println("7. Tim kiem khach hang theo ID");
            System.out.println("8. Xoa khach hang theo ID");
            System.out.println("9. Luu danh sach khach hang hien tai ra file");
            System.out.println("0. Thoat chuong trinh");
            System.out.print("Moi ban chon chuc nang (0-9): ");
            try {
                choice = Integer.parseInt(scanner.nextLine().trim());
            } catch (NumberFormatException e) {
                System.out.println("Vui long nhap so nguyen hop le!");
                continue;
            }

            switch (choice) {
                case 1:
                    bst.inputCustomerList(scanner);
                    break;
                case 2:
                    System.out.println("\n=== DANH SACH KHACH HANG HIEN CO ===");
                    bst.displayAll();
                    break;
                case 3:
                    System.out.println("\n=== THONG KE DO TUOI KHACH HANG ===");
                    bst.calculateAgeStats();
                    break;
                case 4:
                    bst.filterCustomers();
                    break;
```

File: Main.java (Phần 2: Xử lý Case 5 - 0 & Hoàn tất Chương trình)

```
                case 5:
                    bst.sortAndDisplayMaleCustomers();
                    break;
                case 6:
                    bst.findOldestAndYoungestMale();
                    break;
                case 7:
                    System.out.print("\nNhap ID khach hang can tim: ");
                    try {
                        int searchId = Integer.parseInt(scanner.nextLine().trim());
                        Customer found = bst.searchById(searchId);
                        if (found != null) {
                            System.out.println("-> Da tim thay khach hang:");
                            CustomerBST.printTableHeader();
                            found.printRow();
                            CustomerBST.printTableFooter();
                        } else {
                            System.out.println("-> Khong tim thay khach hang co ID: " + searchId);
                        }
                    } catch (NumberFormatException e) {
                        System.out.println("ID phai la so nguyen!");
                    }
                    break;
                case 8:
                    System.out.print("\nNhap ID khach hang can xoa: ");
                    try {
                        int deleteId = Integer.parseInt(scanner.nextLine().trim());
                        boolean deleted = bst.deleteById(deleteId);
                        if (deleted) {
                            System.out.println("-> Da xoa thanh cong khach hang co ID: " + deleteId);
                        } else {
                            System.out.println("-> Khong tim thay khach hang co ID: " + deleteId + " de xoa.");
                        }
                    } catch (NumberFormatException e) {
                        System.out.println("ID phai la so nguyen!");
                    }
                    break;
                case 9:
                    bst.saveToFile(FILE_NAME);
                    break;
                case 0:
                    bst.saveToFile(FILE_NAME);
                    System.out.println("Tam biet! Chuong trinh ket thuc.");
                    break;
                default:
                    System.out.println("Lua chon khong hop le, vui long chon tu 0 den 9.");
            }
        } while (choice != 0);

        scanner.close();
    }
}
// ==================== KẾT THÚC TOÀN BỘ CHƯƠNG TRÌNH ====================
```